

Yung-Han (Tina) Chang

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RESEARCH INTEREST

My research focuses on developing statistical methods for clinical and biomedical data. I am particularly interested in survival analysis, causal inference, clinical trial design and high-dimensional modeling for evaluating treatments predicting patient outcomes and identifying biomarkers. My recent work also explores integrating foundation-model representations with classical biostatistical methods to support risk prediction and precision medicine.

EDUCATION

University of Texas MD Anderson Cancer Center

Houston, TX

Ph.D., Quantitative Science, Biostatistics track

May 2028

University of California, Los Angeles

Los Angeles, CA

M.S., Biostatistics

Jun. 2024

National Taiwan University

Taipei, Taiwan

B.S., Biostatistics and Bioinformatics

Jun. 2020

EXPERIENCE

Graduate Student Researcher

Aug. 2024 - Present

MD Anderson Cancer Center, Department of Biostatistics

Houston, TX

- Building an evaluation harness for a pretrained RNA foundation model that benchmarked 121,620 transcript embeddings using FAISS retrieval and linear probing and identified embedding anisotropy.
- Implementing proximal gradient optimization for high-dimensional RMST prediction with sparse group LASSO regularization and reduced MAE from 1.20 to 1.04.
- Developed a Bayesian joint-modeling framework for survival data, quantifying direct and ctDNA-mediated treatment effects on overall survival in colorectal cancer.
- Developed an empirical Bayes framework to integrate evidence across multiple lung cancer cohorts and improve the robustness of replicated genetic findings while maintaining comparable risk prediction with 87.3% fewer variants. [[Link](#)]
- Developed multivariate predictive models integrating genomic and transcriptomic data across six cancers to identify disease-associated biomarkers and improved discovery of significant associations by 164%. [[Link](#)]

Graduate Student Researcher

Sep. 2022 - Jun. 2024

UCLA, Department of Computational Medicine

Los Angeles, CA

- Designed a factorial Monte Carlo simulation study across 22 loci to evaluate the cross-population transportability of SuSiE and elastic net prediction models, showing that increasing model complexity reduced predictive performance by 62–66%.

Data Scientist

Sep. 2021 - Sep. 2022

National Taiwan University Hospital

Taipei, Taiwan

- Conducted a GWAS of headache in the Taiwan Biobank, identifying genetic factors contributing to disease risk. [[Link](#)]

- Performed a nationwide retrospective cohort study of 20,022 sepsis patients, demonstrating that sepsis is an independent risk factor for in-hospital cardiac arrest. [\[Link\]](#)
- Assisted in predictive modeling and external validation of *FluDeep*, a deep learning system combining chest X-rays with clinical data to assess influenza mortality risk. [\[Link\]](#)

Data Scientist Intern

Oct. 2017 - Aug. 2018

IQVIA

Taipei, Taiwan

- Analyzed pharmaceutical sales and market data and translated findings into client-facing reports and visualizations to support commercial decision-making.
- Performed data validation and quality control for multinational pharmaceutical clients across Asian markets to ensure the accuracy and consistency of analytical deliverables.

CLINICAL TRIAL DESIGN PROJECTS

Phase I–III Trial Design and Simulation

Fall 2025

- **Patient-centered phase I dose finding:** Extended the Bayesian model-averaging CRM to co-primary clinician-assessed DLT and patient-reported DLT (PRO-CTCAE) endpoints via a joint posterior target-interval decision rule; across 2,000 simulated trials per scenario, the bivariate design improved correct MTD selection by up to 16 percentage points and reduced mean DLT events by roughly one third relative to univariate CRM, with operating characteristics robust to endpoint correlation.
- **End-to-end development pathway simulation:** Simulated 10,000 complete phase I–III programs per scenario (BOIN dose finding → Simon two-stage screening with Bayesian toxicity monitoring → event-driven group-sequential OS trial with O’Brien–Fleming spending boundaries), quantifying unconditional probability of success and expected trial duration to support go/no-go and program-level design decisions.

SKILLS

- **Programming & Computing:** R, Python, SAS, Bash, MySQL, Linux, High Performance Computing
- **Version Control:** Git/GitHub

SELECTED PUBLICATIONS

- **Yung-Han Chang**, Jinyoung Byun, Bryan R Gorman, Rayjean J Hung, James D McKay, Christopher I Amos, Saiju Pyarajan, Arjun Bhattacharya, Ryan Sun. [Formal Statistical Replication Analysis in Lung Cancer Genome-Wide Association Studies](#). *medRxiv*, 2025.
- **Yung-Han Chang**, Sean T Bresnahan, S Taylor Head, Tabitha A Harrison, Yao Yu, Chad D Huff, Bogdan Pasaniuc, Sara Lindström, Arjun Bhattacharya. [Isoform-level analyses of 6 cancers uncover extensive genetic risk mechanisms undetected at the gene-level](#). *British Journal of Cancer*, 2025. *Emerging Leaders in Cancer Special Edition*.
- Meng-Han Tsai, Sung-Chu Ko, Amy Huaishiuan Huang, Lorenzo Porta, Cecilia Ferretti, Clarissa Longhi, Wan-Ting Hsu, **Yung-Han Chang**, Jo-Ching Hsiung, Chin-Hua Su, Filippo Galbiati, Chien-Chang Lee. [Predicting mortality in hospitalized influenza patients: integration of deep learning-based chest X-ray severity score \(FluDeep-XR\) and clinical variables](#). *Journal of the American Medical Informatics Association*, 2024
- Ye Liu, Chia-Hung Yo, Jiun-Ruey Hu, Wan-Ting Hsu, Jo-Ching Hsiung, **Yung-Han Chang**, Shyr-Chyr Chen, Chien-Chang Lee. [Sepsis increases the risk of in-hospital cardiac arrest: a population-based](#)

PRESENTATIONS

Bayesian Survival Mediation Analysis with Structured Configuration

- Oral presentation at the Joint Statistical Meetings, Boston, MA, Aug. 2026.

Formal Statistical Replication Analysis in Lung Cancer Genome-Wide Association Studies

- Oral presentation at the Conference on Statistics in Genomics and Genetics, Atlanta, GA, May. 2026.
- Poster presentation at the American Society for Human Genetics (ASHG) Annual Meeting, Boston, MA, Oct. 2025.
- Poster presentation at the International Genetic Epidemiology Society (IGES) Annual Meeting, Cologne, Germany, Aug. 2025.

Isoform-level analyses of 6 cancers uncover genetic risk mechanisms undetected at the gene-level.

- Poster presentation at the ASHG Annual Meeting, Denver, CO, Nov. 2024.

Portability of TWAS Gene Expression Across Ancestries

- Poster presentation at the IGES Annual Meeting, Nashville, TN, Nov. 2023.
- Poster presentation at the ASHG Annual Meeting, Washington DC, Nov. 2023.